Production grade AI code editor

The challenge was moving beyond simple AI chat to create an autonomous agent capable of manipulating a file system and executing code in real-time.

The goal was to architect a seamless IDE experience with agentic work flows, in browser execution, and real-time sync

Made with Slidey

The Vision: Autonomous IDE

agentic workflows

AI that can create, edit, and delete files

Browser-Native Execution

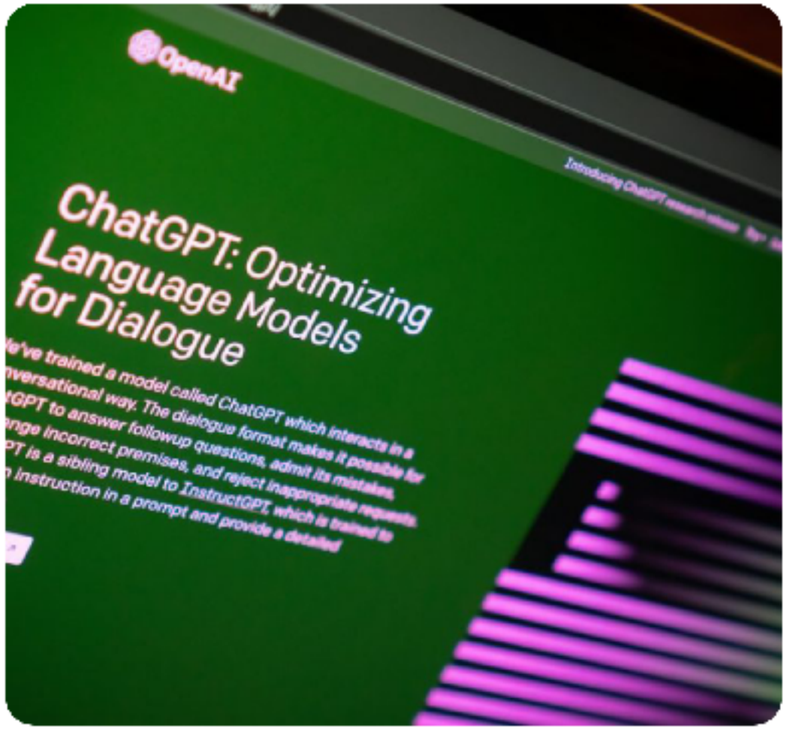
A live preview environment without local setup.

Real-time sync

instance state updates across the file tree and terminal.

Made with Slidey

Core Technologies: Next.js & Typescript

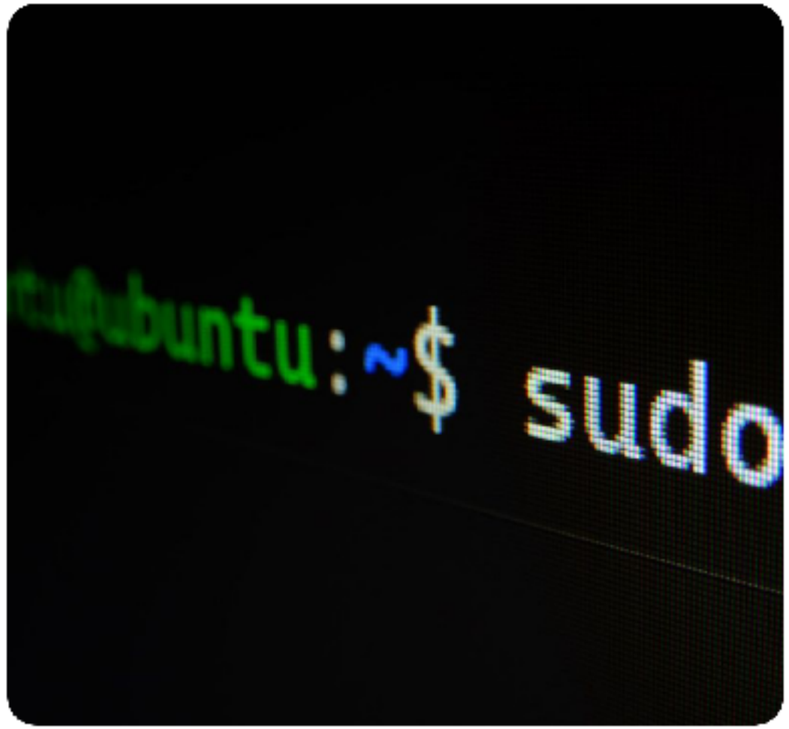


AI orchestration

integrated Claude (anthropic) with inngest to manage long running tool-calling loops for autonomous file management.

The IDE Layer

used code mirror 6 for the editor and web containers to run a full nodejs environment directly in the browser



Context & Data

used firecrawl to scrape live documentation for the AI's context and convex for sub - 100 ms real time databa

Made with Slidey

Building the Agent: Result

1

Core Functionality Explained

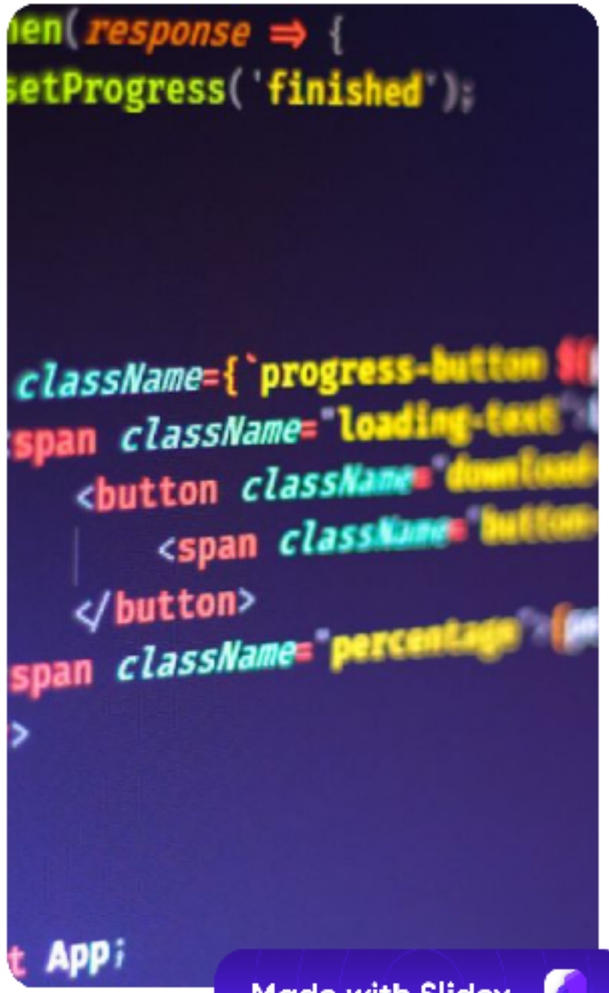
The result is fully functional SAAS IDE. users can prompt and entire app into existence, preview it live, export it to GitHub. By off loading execution to the browser via web containers, I reduced server costs while providing an 'instant-on' developer experience.

Made with Slidey

Future Directions & Conclusion

AI Agent IDE Development:

Built of full stack IDE using nextjs and webcontainers that allows an autonomous Claude-powered agent to write, execute, and deploy code directly from the browser.



Made with Slidey